

## Session 9 (AFRM):

GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model:

Time-varying  $\sigma^2$  or Variance.

GARCH model helps to model time varying variance or volatility by capturing the dynamics of a series.

GARCH model is made of two eqs:-

(a) Mean eq.: It mainly captures the dynamics of return. We mainly use AR (autoregressive) model or MA (moving average) model or ARMA (autoregressive moving average) model to model mean eq. However, mean eq. plays very little role in VAR estimation. Due to this will not model the mean eq.

$$R_{t+1} = \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, \sigma^2)$$

$\rightarrow$  Error term

(b) Variance eq.:

$$\sigma_{t+1}^2 = w + \alpha R_t^2 + \beta \sigma_t^2$$



Here  $w$ ,  $\alpha$ , &  $\beta$  are unknown coefficients & are estimated using Maximum likelihood estimation.

Here if  $w=0$ ,  $\alpha=(1-\lambda)$ ,  $\beta=\lambda$

$$\sigma_{t+1}^2 = 0 + (1-\lambda)R_t^2 + \lambda \sigma_t^2$$

$$\sigma_{t+1}^2 = (1-\lambda)R_t^2 + \lambda \sigma_t^2$$

Note:

$$\sigma^2 = \frac{w}{1-(\alpha+\beta)}$$

$\alpha + \beta < 1$  [just to make long run variance to be finite & positive]

## Steps to implement GARCH model:

1. First take return series.

2. Take initial (wrong) values of

$$W = 0.00005$$

$$\alpha = 0.15$$

$$\beta = 0.75$$

3. First, take sample variance to get the first day variance.

From 2nd day onward, use

$$\sigma_{t+1}^2 = W + \alpha R_t^2 + \beta \sigma_t^2$$

to estimate variance for all the given days.

3. Next, we need to estimate the likelihood that given return observation is coming from the given assumed distribution (e.g. Normal distribution) with mean zero & variance as GARCH variance. The likelihood function ( $L_i$ ) is a pdf [prob. density function] of given distribution (e.g. Normal distribution)

$L_i$  = pdf of normal distribution for return

$$L_i = \frac{1}{\sqrt{2\pi}\sigma_i} \cdot \exp\left(-\frac{(r_i - \mu)^2}{2\sigma_i^2}\right) = \text{pdf of Normal distribution}$$

$$L_i = \text{Norm.Dist}(r_i, 0, \sigma_i, \text{False})$$

→ for pdf

→ Estimate likelihood function of each return observation.

④ Estimate log-likelihood function (LLF<sub>i</sub>)  
 $LLF_i = \ln(L_i)$

⑤ Estimate joint log-likelihood function  
(idea is to know the joint likelihood of returns observations coming from same normal distribution with zero mean & known variance).

$$\text{Joint } LLF_i = \sum_{i=1}^n LLF_i = LLF_1 + LLF_2 + \dots + LLF_n$$

⑥ Next, maximize the joint LLF & estimate  $w, \alpha$  &  $\beta$ .

→ Once we obtain optimized  $w, \alpha$  &  $\beta$ , we get optimized conditional variance which we can use for estimating & backtesting VAR.

⑦ Estimate forecasted variance.

For long position,

$$95\% \quad 1\text{-day } VAR_{t+1} = -1.645 * \sigma_{t+1} * VP_{t+1}$$

$$99\% \quad 1\text{-day } VAR_{t+1} = -2.33 * \sigma_{t+1} * VP_{t+1}$$

Note:

$$10\text{-day } VAR. = 1\text{-day } VAR * \sqrt{10}$$

## Introduction to R:

### ① Clear the memory:

```
rm(list=ls())
```

### ② Setting working directory: To set working directory, use `setwd("path")`

### ③ Importing data from CSV file in R:

Use `read.csv` command

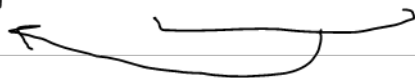
```
data = read.csv("Filename.csv", header=T)  
any name
```

→ The data imported using `read.csv` command is imported as data frame.

### ④ Separating data variable for analysis:

Way 1: If imported data is in the form of data frame, we can separate or particular series by referring that series.

```
nifty = data$Nifty
```



Way 2: The imported data can also be considered as matrix.

$$\begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$$

- Mostly data taken from different databases, the data will be in different columns.
- We can extract the data present in a column(s) by referring the given column(s).

datavariable [row, column]

- If we want to extract data in a given column, then mainly mention the column & keep row part empty.

e.g. to extract data present in 2nd column, use

datavariable [, 2]

- To extract data present in a row, mention the row & keep the column empty.

→ e.g. to extract data in 3rd row, use

data [3, ]

- We can also refer a data point by referring row & column.

data [5, 2] = provide data value at row 5 & col 2.

- We can also extract range of data e.g. take data present in rows 1 to 250 & column 2.

data [1:250, 2]  
rows col.

⑤ length of data series:

$$n = \text{length}(\text{variable})$$

⑥ Estimating return:

Simple return,

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{\text{First difference of prices}}{\text{lagged price}} = \frac{\text{diff}(\text{prices})}{\text{lag}(\text{price})}$$

Continuously compounded return:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln(P_t) - \ln(P_{t-1}) = \text{First difference of } \ln(\text{prices}) = \text{diff}(\ln(\text{price}))$$

⑦ Plotting: To plot, use  
`plot(series, type="l")`

⑧ Combining different variables in different columns: use `cbind` command.

`output variable = cbind(var1, var2, ---)`

⑨ Saving results in csv file:

write.csv (name of variable, "Filename.csv")  
to save

⑩ Conditional Statement:

① if - else -

Syntax:

```
if (condition) {
```

```
  - - - - -
```

```
} else {
```

```
  - - - - -
```

```
}
```

if condition is true, then  
codes given here are  
executed

if condition is false,  
then codes given here  
are executed.

e.g.:  $x = \text{rnorm}(1, 0, 1)$

```
if (x > 0) {
```

```
  y = 5 * x
```

```
} else {
```

```
  y = 0
```

```
}
```

② if we have multiple conditions to  
test.

use if - else if -

Syntax:

```
if (Condition1) {  
  - - - - -  
  - - - - -  
}
```

if condition 1 is true, then codes given here are executed

```
} else if (Condition2) {  
  - - - - -  
  - - - - -  
}
```

if condition 2 is true, then codes given here are executed.

```
} else {  
  - - - - -  
  - - - - -  
}
```

if conditions 1 & 2 are false, then codes given here are executed.

⑪ Loop: To repeat same task again and again.

Syntax:

```
for ( i in start index : end index ) {  
  - - - - -  
  - - - - -  
  - - - - -  
}
```

index variable

e.g. generate 1st ten multiples of 5.

```
y=0  
for ( i in 1:10 ) {  
  y = 5 * i  
}
```



## Implementing GARCH model in R:

To GARCH model in R, we will use "rugarch" package.

